BEM 103

Class 5:
Markets and Exchanges

Assets exit so they have prices all the time

Last bit of riskless bond

Exchange as measure of value

Secondary markets and the double auction

Primary markets and underwriting

Pricing bonds, the last step

- Lets consider 5 year bonds with a coupon c

1-Jan-27	1-Jan-28	1-Jan-29	1-Jan-30	1-Jan-31
c	c	c	c	100+c

- Claim: if r= c the price of the bond is 100 on January 2
  - Consider January 2 2030.
    - Invest 100 get 100+r or buy bond for p get 100+c
    - Return to bond $\frac{100+c}{p}$ return to alternative $\frac{100+c}{100}$
    - no arbitrage => p=c
- Now step back to Jan 2 2029.
  - Invest 100 get 100+r or buy bond at price p today and sell in one year for 100 and get c

What if r and c are different

- Claim if r>c then bond must sell at a discount
  - Consider January 2 2030.
  - Invest 100 get 100+r or buy bond for p get 100+c
  - Return to bond $\frac{100+c}{p}$ return to alternative $\frac{100+r}{100}$
- No arbitrage requires

  $\frac{100+c}{p} = \frac{100+r}{100}$ or $\frac{p}{100} = \frac{100+c}{100+r}$ or $\mathrm{p} = 100\frac{100+c}{100+r}$
- Now you know the price on Jan 2 2030
  - So you can compute the no arbitrage condition on Jan 2 2029…and keep going backwards

Price of bond with 3 % coupon T years before redemption depending on the interest rate

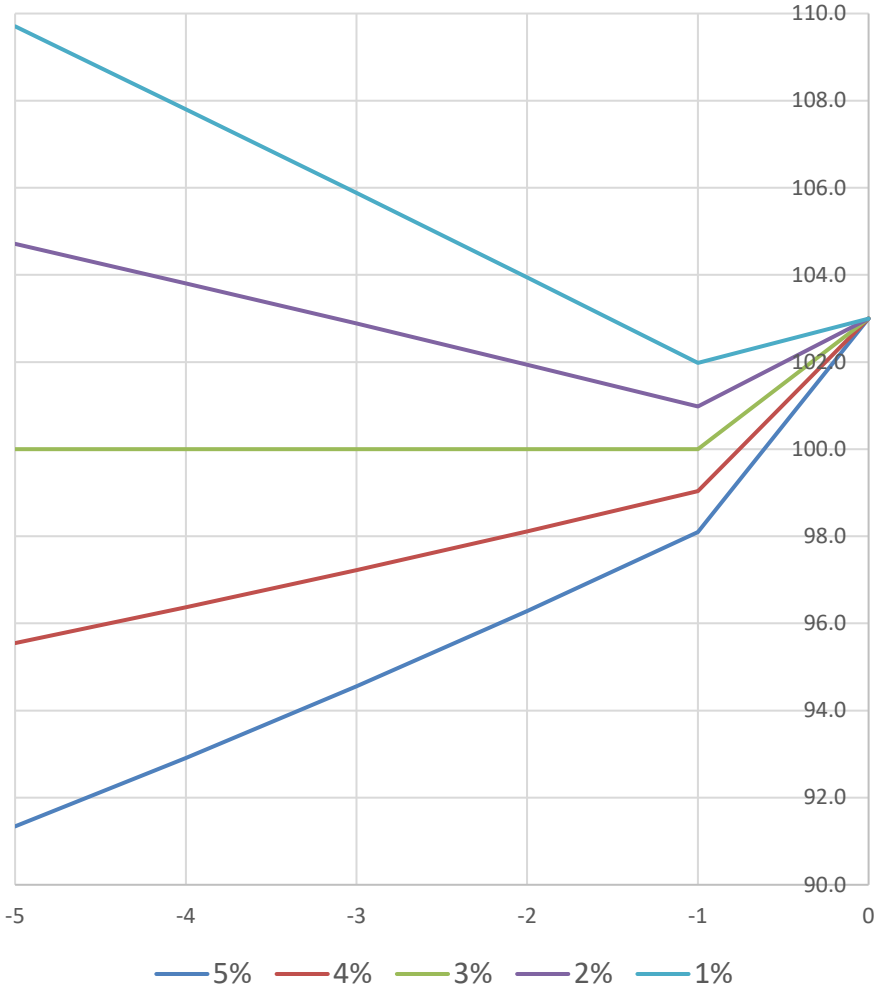
	5%	4%	3%	2%	1%
-5	91.3	95.5	100	104.7	109.7
-4	92.9	96.4	100	103.8	107.8
-3	94.6	97.2	100	102.9	105.9
-2	96.3	98.1	100	101.9	103.9
-1	98.1	99.0	100	101.0	102.0
-1 day	103	103	103	103	103

Price of bonds with different coupons T years before redemption when the interest rate is 3%

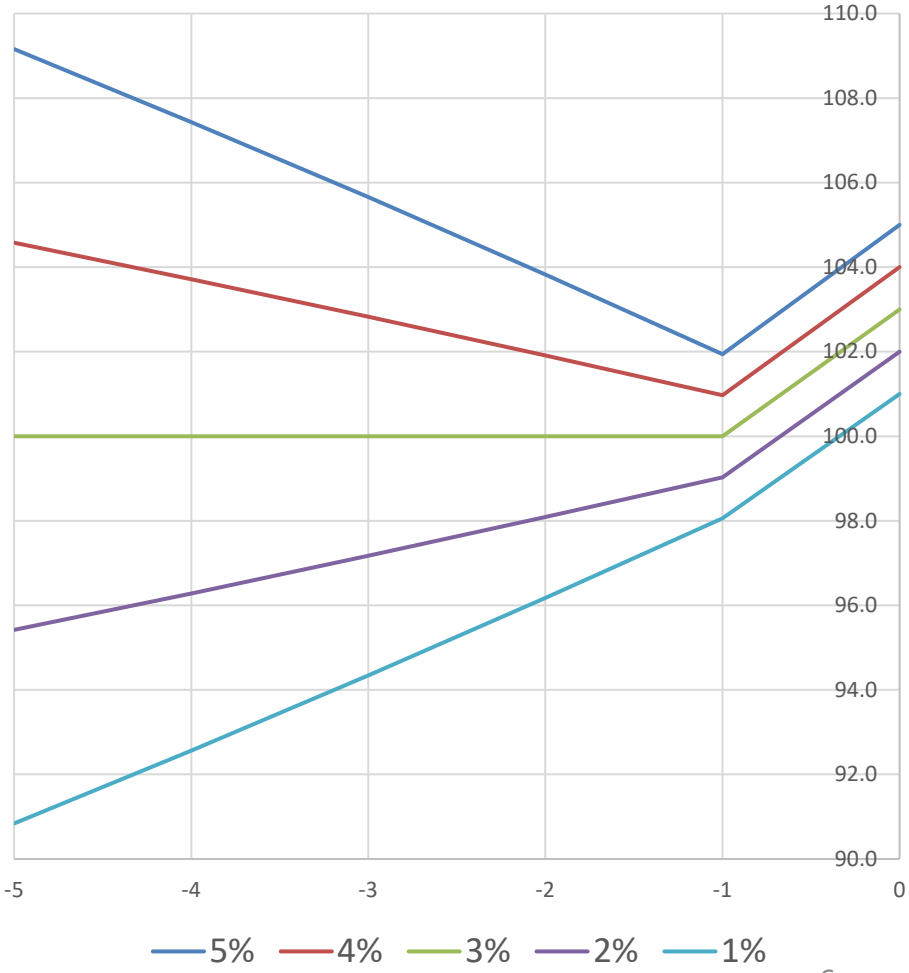
	5%	4%	3%	2%	1%
-5	109.2	104.6	100	95.4	90.8
-4	107.4	103.7	100	96.3	92.6
-3	105.7	102.8	100	97.2	94.3
-2	103.8	101.9	100	98.1	96.2
-1	101.9	101.0	100	99.0	98.1
-1 day	105	104	103	102	101

Same pb in pictures

Price of 3% bond under different interest rates



Price bonds with different coupons with 3% interest rates



Quiz time

- No computers no phone
- Just your mind + paper+pen/pencil

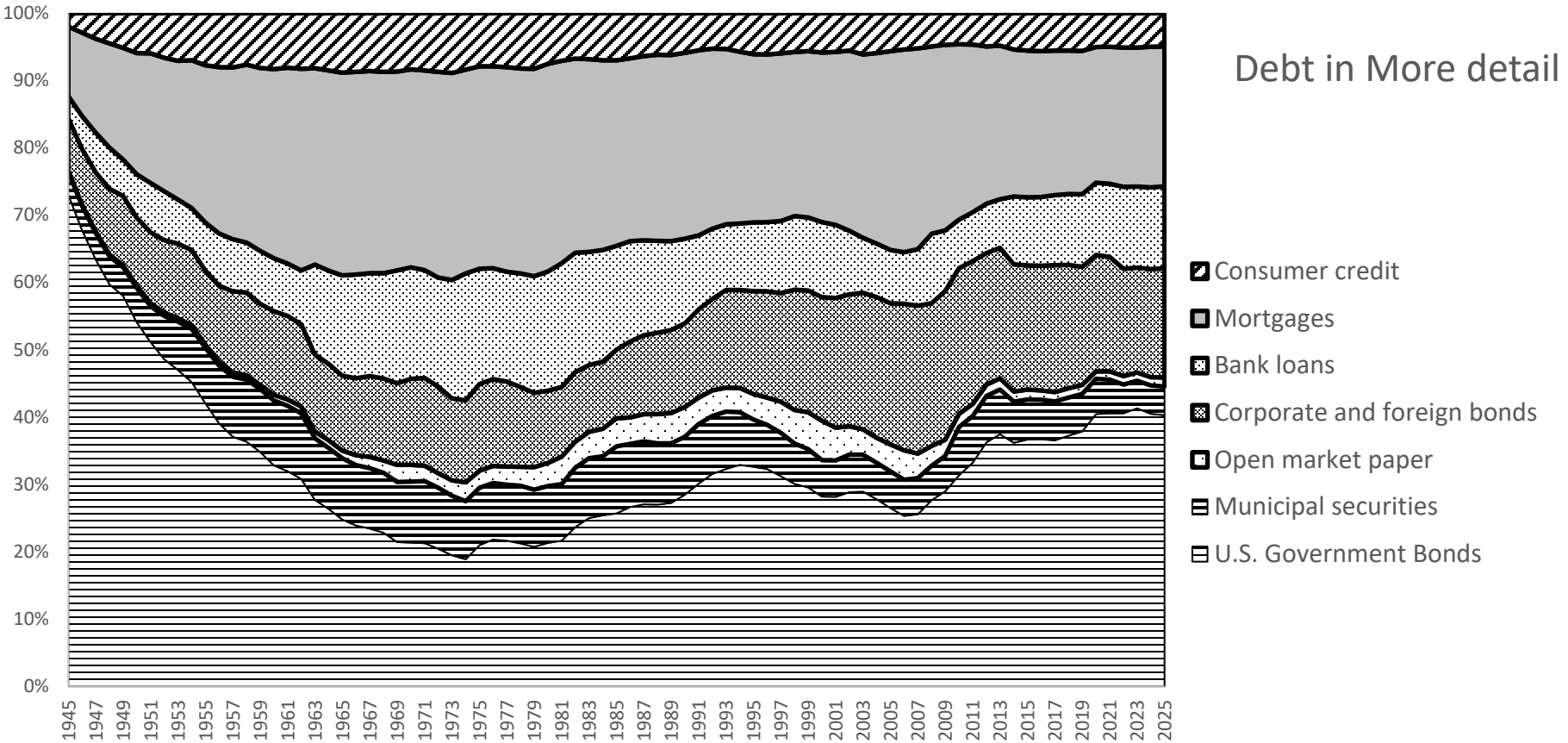
Markets and Echanges
Key concepts

- Banks vs markets
- Liquidity
  - The ability to sell
- Double Auction
  - Bids and ask
  - Bid Ask spread
- Primary and secondary markets
- Intermediaries and transformation

Demand for liquidity

- Liquidity is the ease with which you can sell (buy) an asset
  - If an asset is perfectly liquid you can sell it instantly at the last observed price
  - Related to competition
- Back to term structure of interest rate
  - Rather hold long securities because of the term premium
  - But may face a liquidity issue when need cash
- Liquidity is endogenous
  - Many assets are not tradable, you have to wait until they mature
  - Some are tradable but not traded so you have to find a counterparty
  - Some are publicly traded, so it is ‘easy’ though there may be a price issue
  - Liquidity is partly total value of asset (e.g. number of shares)

Bonds and Debts



- Debt Common
- Traded debt common but a fraction of total
- Why such a large fraction of all debt is government

Side bar
Intermediaries and transformation

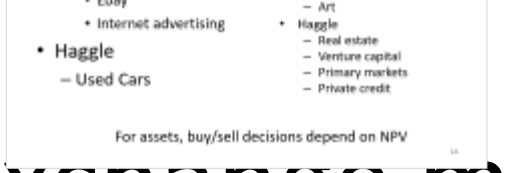
- Capital markets
  - Peer to peer
    - Borrowers find their lenders or vice versa
    - Entrepreneurs find their investors
  - Brokered
    - Hire an agent (broker) to find a counter party
    - Advantages and problem
      - https://www.newsweek.com/nicolas-puech-hermes-switzerland-geneva-court-case-1932131
    - Agent is not a party to the transaction
  - Transformation
    - Investor /lender buys a security from intermediary
    - Intermediary buys a different security from entrepreneur/borrower
    - Banks hedge funds ETFs

Liquidity History

- <u>Assets have been around for a long time</u>
  - Equity nor bonds were tradable easily.
  - Equity in specific assets (so you sold assets rather than securities) or in partnerships (where you can't sell your position)
  - Debt contracts were peer to peer
    - Could not be transferred without the agreement of the debtor)

Innovation

- Joint stock legal form
  - Corporation
- Commercial paper
  - Letter of exchange (endorsement)
- Transferable bonds
- For stock and bonds, need a recording system
  - At the company (so it can pay interest and dividends)
- Assets that can be traded require trading venues



Exchange mechanisms

- Goods and services
  - Posted price
    - Stores
  - Auction
    - Ebay
    - Internet advertising
- Haggle
  - Used Cars

For Assets

- Posted price
  - Bank deposits
  - Certificates of deposit
  - Mortgages
- Auction
  - Public markets
  - Art
- Haggle
  - Real estate
  - Venture capital
  - Primary markets
  - Private credit

For assets, buy/sell decisions depend on NPV

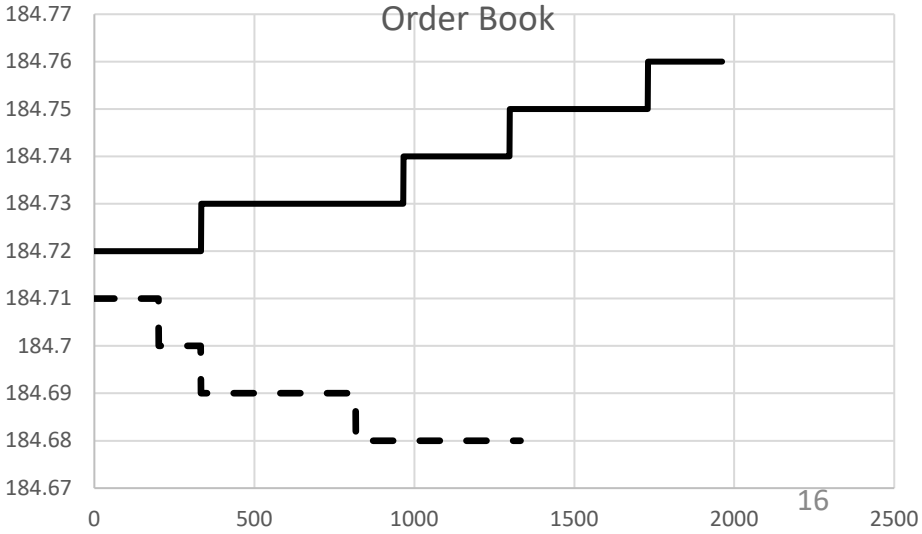
Organized exchanges

- Date from the 17th century
  - Changes to create more liquid securities
    - Issuers (governments realize that investors are willing to pay for liquidity)
  - Public exchanges (listed)
    - NYSE
    - Involves some sort of auction
    - Order book
  - Dealers/broker
    - Over the counter-OTC
    - Contact the dealer broker with an order they find the counterparty.
  - Nasdaq is somewhere in between

Classic Exchange

13:14:01 Tuesday						
		Book		Trades		
	Quantity	Price		Price	Quantity	Time
Asks	233	184.76		184.73	100	13:14:01
	432	184.75		184.73	100	13:14:01
	332	184.74		184.74	100	13:13:56
	632	184.73		184.73	200	13:13:56
	333	184.72		184.71	85	13:13:50
Bids	200	184.71		184.71	2	13:13:50
	132	184.7		184.73	95	13:13:41
	232	184.69		184.72	100	13:13:41
	252	184.68		184.72	100	13:13:41
	516	184.67		184.72	1	13:13:41

https://www.cboe.com/us/equities/market_statistics/book/aapl/



Exchanges not everything

- Lots of privately held companies
  - Cargill, Koch Industries, and Mars (candy), SpaceX
  - Start ups
  - And the mom and pop store
- Today
  - Most firms are privately held (no secondary market for equity)
  - Most of the capital (firm value) is in public companies.
- With debt things are a bit more complicated
  - Because of banks and other financial intermediaries
  - Private debt is mostly non market.

Goals of Secondary Markets

- Liquidity:
  - Aggregate disparate sources of demand and supply
- Transparency:
  - All important actors can share information before making decisions
- Mechanism is auction.
  - Initially one-sided, sellers offer item with reserve price, and buyers bid
  - Soon enough (19th century), continuous double auction
    - Until the 2000s individual investors could not see what current bids looked like, only dealers and brokers.
    - Now IT has made it possible for everyone to see the current bids
- Investors send in orders, transaction occurs when either
  - New bid (offer to buy) exceeds the lowest current ask (offer to sell)
  - New ask (offer to sell) is less than the highest current bid (offer to buy
  - price determined by what was on offer
  - parties pay commission, brokers pay exchange fees)

Types of orders

- Bid/Ask in the double auction takes the form of an order
- The request to an intermediary to buy
  - Market order (X shares, X$ of bonds) is executed at best available price
  - Limit order (buy (sell) X shares at no more (or no less) than price p.
- More complicated orders
  - Stop orders these orders come into effect once the price crosses the stop price.
  - Fill or Kill (do it all or not at all right away)
  - All or None (do it at once or not at all)

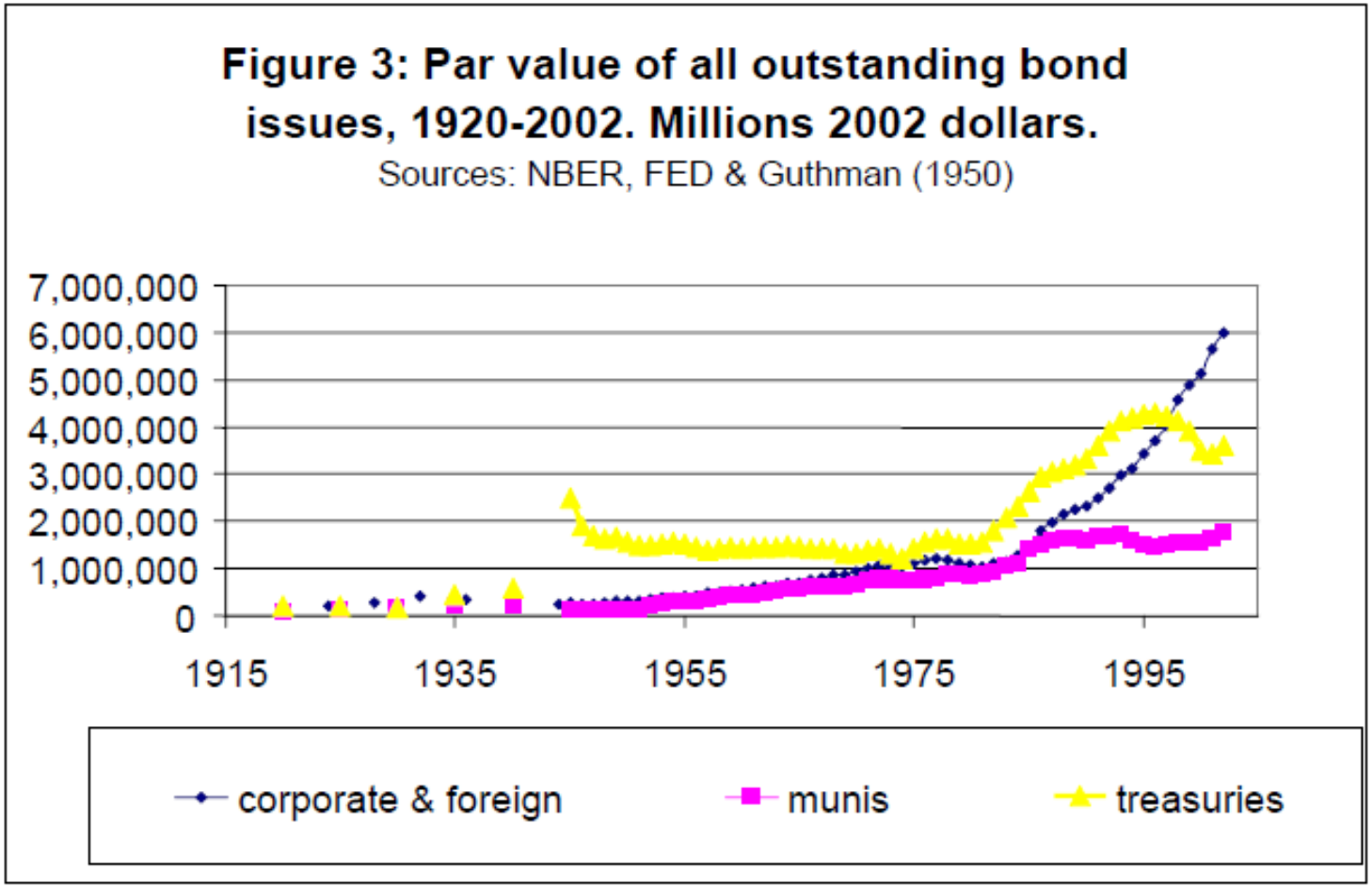
Types of markets

- Two basic technology depending on the function of the agent
  - Broker vs Dealer
- Used to be a big deal (NYSE broker, NASDAQ dealer)
- Now most exchanges are in effect dealers (because ultimate buyer sees the order book)
- Broker: the agent simply reports bids to a specialist (another agent) the specialist makes the market in a stock by matching bids.
- Dealer: the Asker's (seller's) agent 'buys' the security from his principal (the seller) and then transfers it to the bidder's agent who then 'sells' the security to the bidder "buyer."

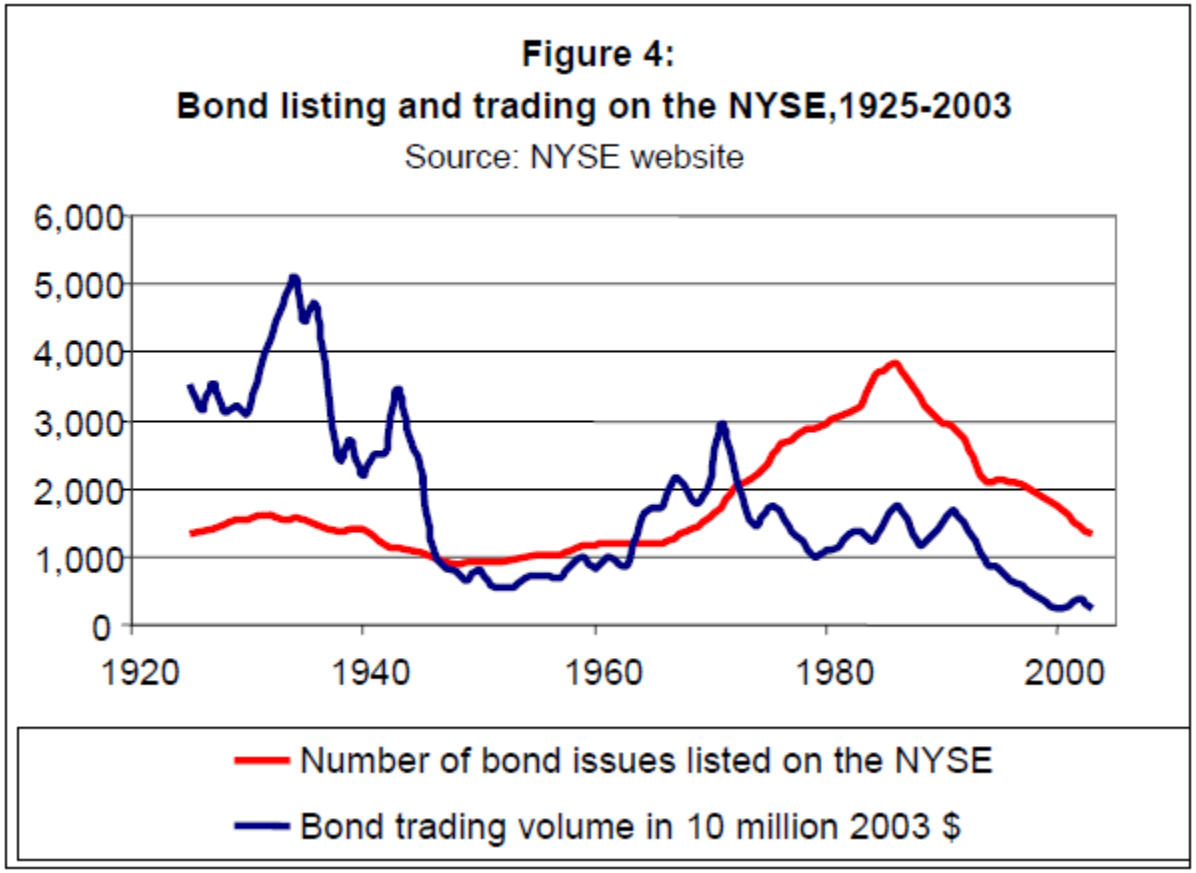
Secondary markets

- Provide liquidity and valuation
- Use continuous double auction
- Increasing returns because investors prefer
  - Low spreads
  - High liquidity
- Issue of fees has been left to the side (but you might consider the effective spread as the bid ask spread + fees)

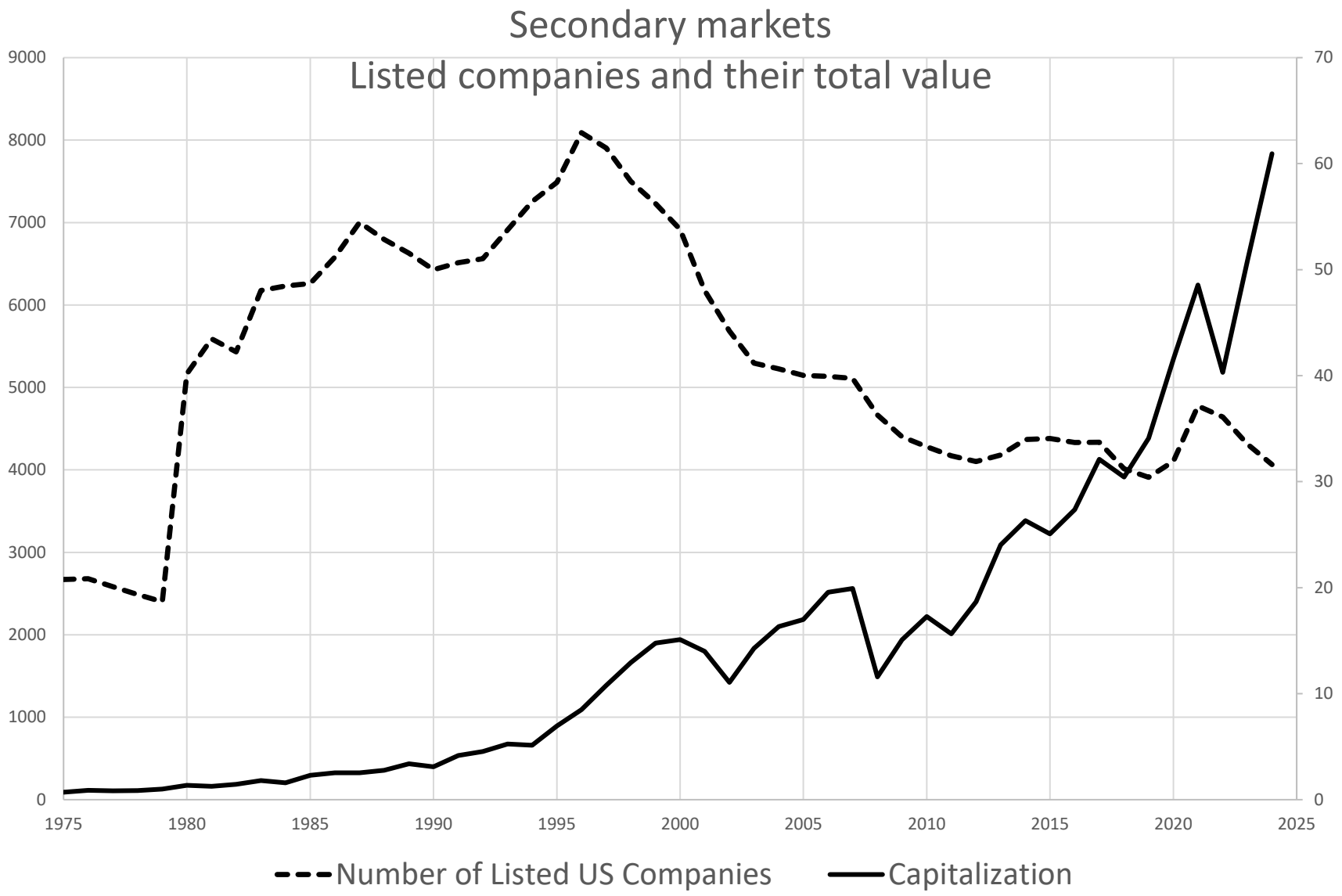
Why is it so hard to find publicly listed bonds?



- The Microstructure of the Bond Market in the 20th Century (Bruno Biais and Richard Green)



- Bonds have overwhelmingly moved to OTC market Why?
- A bit of a mystery, combination of path dependence, rise of institutional investors, and preference by individual investors for holding bond funds over individual bonds
- Pb this is also true for stocks

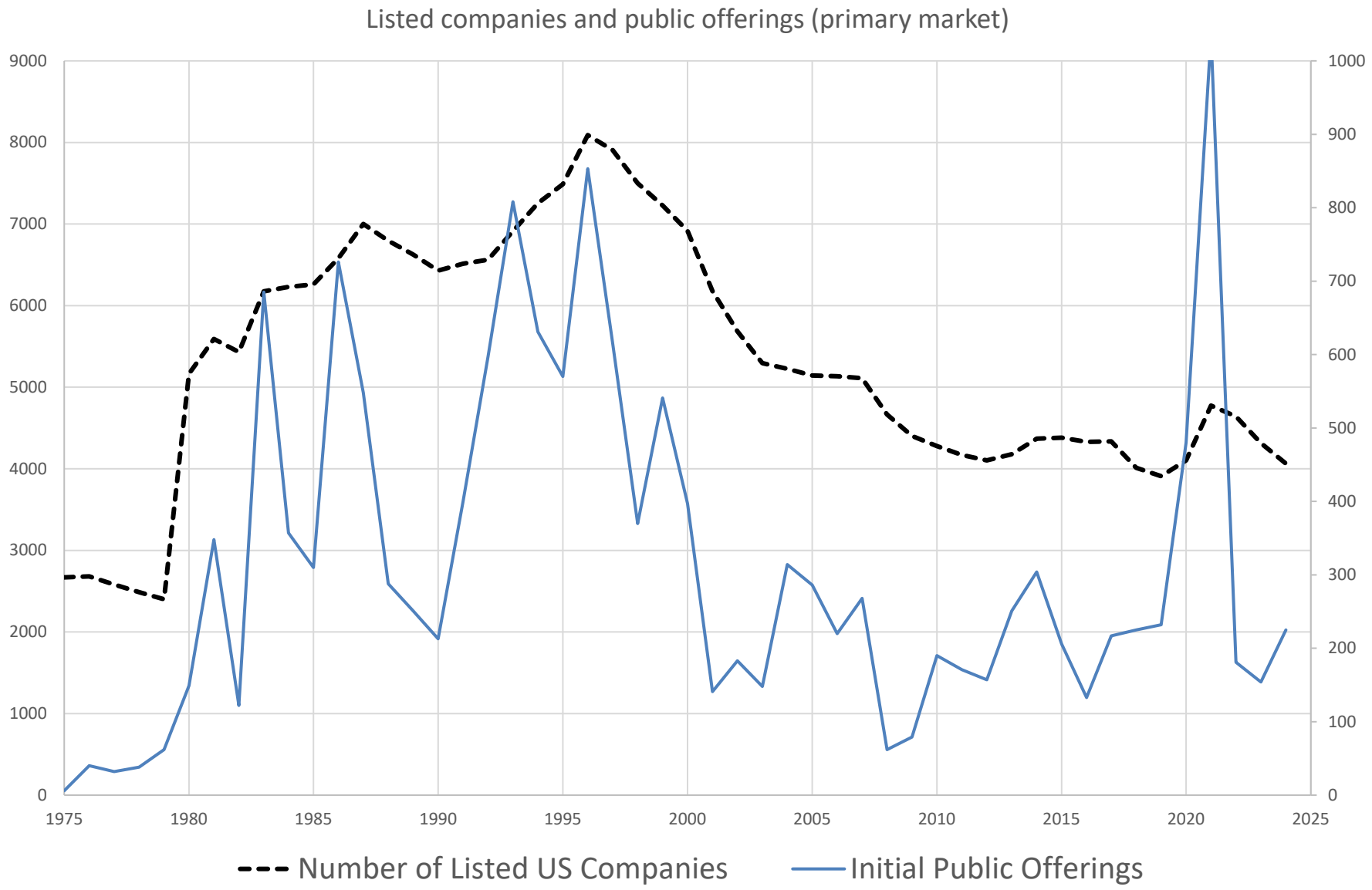

Secondary markets
Listed companies and their total value
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
70
60
50
40
30
20
10
0
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Number of Listed US Companies
Capitalization

Wrap Up

- Two ways to decide value
  - Know the cash flows (vector of x^t) and have an opportunity cost of capital (r) do NPV
  - Observe prices
  - If markets efficient then p=NPV so no arbitrage
  - If markets not efficient then profit opportunities
- Rest of slides are additional material on primary markets

From secondary market to primary market

- Ways to create new securities
  - Private placement
    - With trading restrictions (owners can't resell)—eg venture capital
    - With OTC trading
  - Initial (or secondary) public offering
    - By subscription
    - With an underwriting
- Securities law regulates *any* security that is offered to the public
  - Requirements limited before 1900 everywhere and have grown more important since then with major inflection in U.S. with creation of Securities and Exchange commission
  - Goal is to reduce the likelihood that investors will be taken advantage of by issuers or by intermediaries
  - So two key distinctions. Trade or no Trade, IPO or no IPO
  - Key pt is that Public Offering raise costs to firms
- Focus on IPO



Source https://stockanalysis.com/ipos/statistics/
World Bank,

Process

- Firm decides
  - To issue new securities Hires an investment bank who
  - Should do some due diligence about the firms future revenue prospects and current balance sheet (value the firm—in a NPV way)
  - Find some buyers (build a book)

Underwriting

- Left undecided is IPO price.
- Investment banks have complicated incentives
  - They are hired by the issuer and so to attract future issuers they would like to have 'high' IPO prices
  - Issuers want their money quickly so they would like the investment bank to find buyers for the issue quickly. So price is not everything.
  - They have to place the assets and their life is made easier in this respect if IPO prices are "low."
- Solution is underwriting
  - Becomes common in the 19th century
  - Investment Bank sets price and "buys" all the shares at the IPO price and it gets a commission proportional to the value the issue. The more shares it places with the public the faster it can free up is capital to do another IPO.

Underwriting

- Underwriting is in fact an insurance contract:
  - if the issue is over subscribed, the investment bank is just a pass through.
  - If the issue is undersubscribed, the investment banks has to keep the balance of the shares
- This is an expensive contract
  - In the US IPO fees are about 7% of issue value for firms issuing less than $500 Million and about 5.1% for issues above that
  - In Europe IPO fees are about 4% of issue value for firms issuing less than $500 Million and about 2.5% for issues above that
  - See: Abrahamson, Jenkinson and Jones, "Why don't US Issuers Demand European Fees for IPOs?" Journal of Finance ,forthcoming

Underwriters

- Very concentrated, very profitable business
  - Issues are almost systematically over subscribed (so the insurance is ‘never’ paid).
- The same 8 firms received 81% of all IPO proceeds in the US and 63% of all IPO business in Europe.
- Expertise?
  - Doing these deals is complicated and there are not that many (200 a year or so in US and 100 or so in Europe) so why disperse the expertise
- Reputation?
  - Both sides (issuers and investors) know less about the process than the intermediary. The only way the process can be sustained is if reputation matters, otherwise intermediary will take advantage of one side or the other
- Market power?

Private placement

- The New York Times Company said Monday it had reached an agreement with the Mexican billionaire Carlos Slim Helú for a $250 million loan intended to help the newspaper company finance its businesses…Mr. Slim, who already owns 6.9 percent of the Times Company, would invest $250 million in the form of six-year notes with warrants that are convertible into common shares,[…]The notes also carry a 14 percent interest rate, with 11 percent paid in cash and 3 percent in additional bonds. (NYT January 14 2009)
  - Things to notice, NYT a publicly traded company gets financing from a private source (mostly bonds) but Slim also gets the rights to buy additional shares of common stock
- The New York Times Company said Wednesday that it would pay back the $250 million loan from Carlos Slim Helú, a Mexican telecommunications billionaire, on Aug. 15, freeing itself from one of its larger financial obligations (NYT July 13 2011).
  - The reimbursement is early (and there is a prepayment penalty). But the NY Times.
- Why do this at all?